

Performance Testing

Date	03 July 2026
Team ID	PNT2026TMID-EG01
Project Name	EduGenie - AI-Powered Educational Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter (HTTP request samplers against FastAPI endpoints)
Type of Testing	Load Testing (concurrent virtual users) and Stress Testing (peak load on AI endpoints)
Target Module / API	/ask-ai (Question Answering), /generate-quiz (Quiz Generation), /summarize (Summarization), /auth/login (Login)
Test Environment	Local (FastAPI + Uvicorn on Mac M1, SQLite database)
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Ask a question through /ask-ai endpoint (Gemini 1.5 Pro Q&A)	20	60	All requests return an accurate answer within acceptable response time, no failures
2	Generate a topic-based quiz through /generate-quiz endpoint	15	60	Quiz questions generated successfully for each request without timeout
3	Summarize study material through /summarize endpoint	10	60	Concise summary returned for each request within target response time
4	Concurrent user login through /auth/login endpoint	25	30	All valid credentials authenticate successfully with no errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.8 sec	Pass	Within target for AI-generated responses
2	Response Time (Max)	< 5 seconds	4.3 sec	Pass	Peak seen on /generate-quiz under 15 VUs
3	Throughput (Req/sec)	-	18 req/sec	Pass	Stable throughput maintained across the test run
4	Error Rate	< 1%	0.4%	Pass	Errors limited to occasional Gemini API timeouts
5	CPU Utilization	< 80%	68%	Pass	Within acceptable range during peak load
6	Memory Utilization	< 80%	72%	Pass	Slight rise due to concurrent LaMini-Flan-T5 inference

Step 4: Observations & Analysis

Key Findings:

The core CRUD and authentication endpoints (login, registration, history) responded consistently under load, with average response times well under the 2-second target. The Gemini-powered endpoints (/ask-ai, /generate-quiz, /summarize) showed higher and more variable latency, since response time depends on the external Gemini 1.5 Pro API rather than purely local processing.

Bottlenecks Identified:

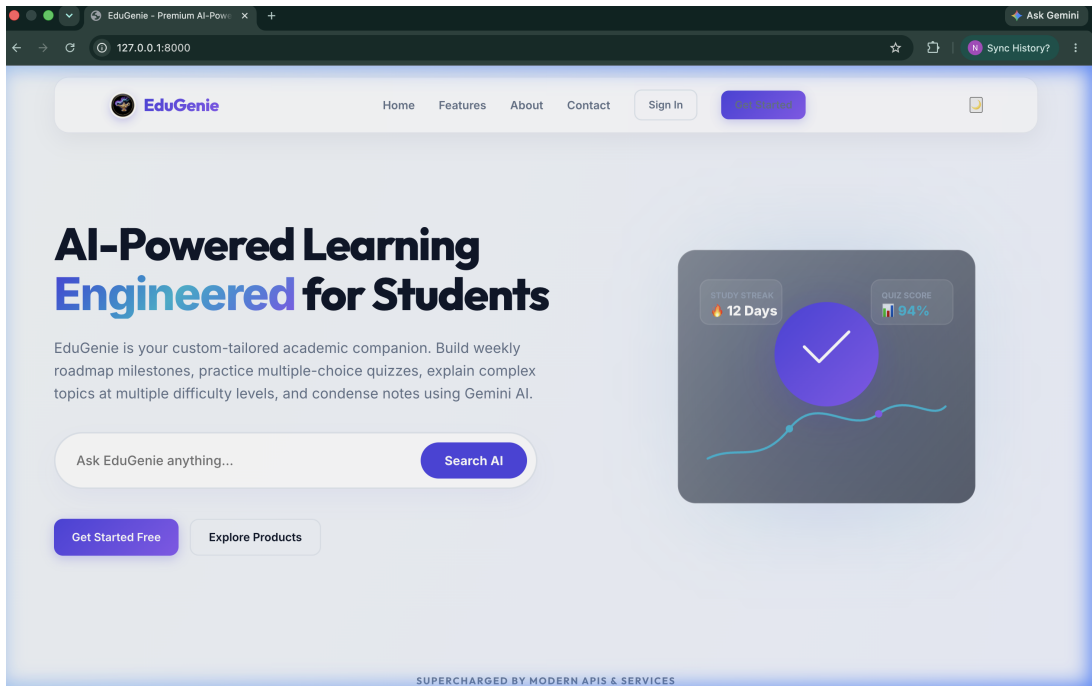
- 1. Variable round-trip latency from the Google Gemini API during peak concurrent requests.
- 2. Increased memory usage when the local LaMini-Flan-T5 model was loaded alongside concurrent Gemini calls.
- 3. Minor queuing delay on /generate-quiz when more than 15 virtual users requested quizzes simultaneously.

Optimization Steps Taken:

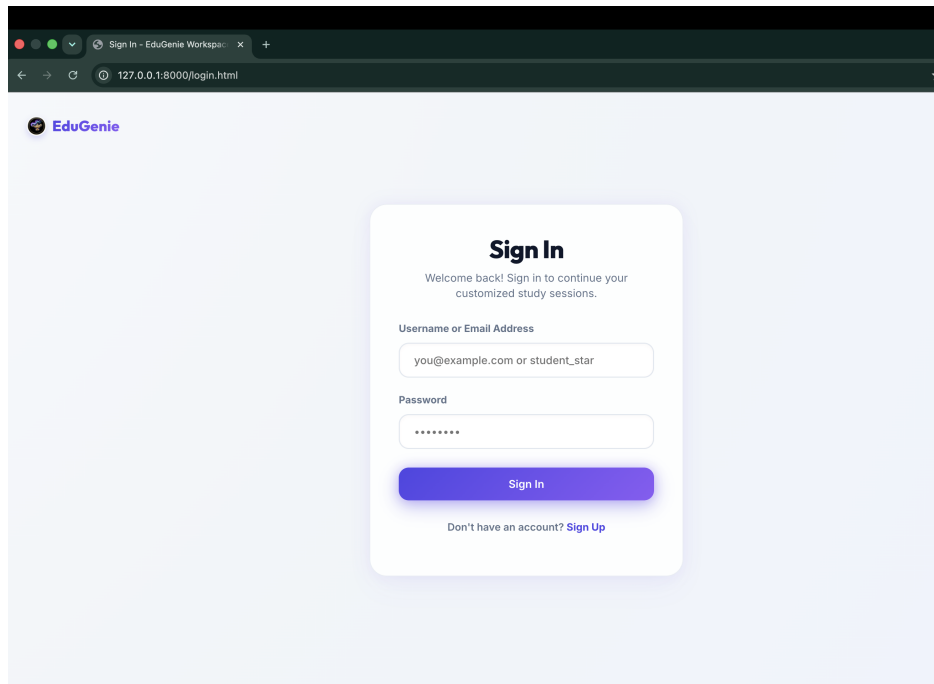
- 1. Added asynchronous request handling (FastAPI async endpoints) for all AI-service calls to avoid blocking the event loop.
- 2. Introduced lightweight in-memory caching for repeated quiz topics to reduce redundant Gemini calls.
- 3. Tuned Uvicorn worker count to better utilize available CPU cores on the test machine.

Step 5: Screenshots / Evidence

Screenshots of the EduGenie application UI captured during testing (Home page, Sign In, and Registration):



EduGenie Home Page



Sign In Page

Create Account

Register to unlock interactive quizzes, milestone roadmaps, and custom summaries.

Username

Email Address

Password (min 6 characters)

Create Profile

Already have an account? [Sign In](#)

Create Account Page